

Football Analytics AI

Cycle 3 — Player Injury Risk Prediction

Documentation/Findings

1. OVERVIEW

1.1 What Cycle 3 Does

Cycle 3 is the most clinically significant component of the project. Where Cycle 1 predicted match outcomes and Cycle 2 estimated shot quality, Cycle 3 turns the analytical lens onto the players themselves. Its goal is to predict, before a season begins, whether a player is at high risk of spending a significant amount of time injured during that season.

The 28-day threshold is deliberately chosen: it corresponds to approximately four weeks, the widely accepted clinical boundary between a minor/moderate injury and a significant injury in the sports medicine literature. An injury at or above this threshold typically involves structural damage — ligament tears, muscle ruptures, stress fractures, or surgeries — with material impact on training and availability.

The cycle spans seven notebooks: Exploration → Preprocessing → Modelling (Random) → Tuning (Random) → Modelling (Chronological) → Tuning (Chronological) → Explainability.

1.2 The ML Problem Type: Binary Classification

- Class 1: High Injury Risk — player predicted to spend ≥ 28 days injured (70.2% of dataset)
- Class 0: Low Injury Risk — player predicted to spend < 28 days injured (29.8%)

Class imbalance: 70/30. A Dummy predicting 'High Injury' always achieves 70.11% accuracy — the accuracy trap. AUC-ROC is the primary metric.

1.3 Why Injury Prediction Matters

- Financial cost: Premier League clubs lose £50–250M per season in wages to injured unavailable players.
- Tactical cost: A long-term injury to a key player destabilises a system for months. Proactive risk identification enables tactical contingencies.
- Player welfare: Chronic injuries shorten careers and affect post-retirement health. Data-driven load management is an ethical obligation.
- Transfer market: Clubs spending hundreds of millions need injury risk as part of acquisition due diligence.

2. AIM AND OBJECTIVES

Aim

To build a robust, temporally valid machine learning pipeline that predicts pre-season binary injury risk for Premier League players using only features knowable before the season starts, evaluated on a held-out future season to simulate real-world deployment.

Objectives

- Conduct thorough EDA to understand injury severity distributions, class imbalance, demographic patterns, and feature-target relationships.
- Design a preprocessing pipeline that constructs the 28-day binary target, removes post-season leakage features, and handles two structurally distinct types of missing values.
- Train and compare five models under both random and chronological splits, quantifying the temporal splitting effect.
- Explain why Cycle 3 shows the opposite temporal pattern from Cycle 1 (chronological split improves rather than reduces performance).
- Optimise the best model via RandomizedSearchCV under chronological conditions, producing the final risk-scoring model.
- Interpret predictions using SHAP to validate medically coherent patterns and enable per-player explanations.

3. DATASET DESCRIPTION

3.1 Source

Source: data/raw/player_injuries.csv — 1,301 rows × 30 columns representing individual player-season records for Premier League players across 5 seasons: 2016–2020. Each row is one player-season combination. 604 unique players are represented. A player active across all 5 seasons appears 5 times in the dataset.

3.2 Target Variable

```
High_Injury = (season_days_injured >= 28).astype(int)
```

Value	Meaning	Count	Proportion
1	High Injury (≥ 28 days)	913	70.2%
0	Low Injury (< 28 days)	388	29.8%

Injury distribution breakdown: No injury (0 days) ~5%, Minor (1–6 days) ~12%, Moderate (7–27 days) ~12%, High (28–89 days) ~28%, Severe (90+ days) ~43%. The 28-day threshold is the sports medicine standard separating moderate from significant injuries.

3.3 Features: Pre-Season vs Leakage

Group 1 — Physical/Biographical Attributes:

Feature	Description
age	Player's age at season start
height_cm	Height in centimetres (FIFA data)
weight_kg	Weight in kilograms
bmi	Body Mass Index (weight / height ²)
pace	FIFA pace rating
physic	FIFA physicality rating
fifa_rating	Overall FIFA ability rating
position_numeric	Ordinal: GK=0, DEF=1, FWD=2, MID=3
work_rate_numeric	Ordinal work rate score (2.5–4.0)

Group 2 — Career Injury History:

Feature	Description
avg_days_injured_prev_seasons	Average injury days per prior season
season_days_injured_prev_season	Injury days in immediately preceding season
significant_injury_prev_season	Binary: significant injury (≥28 days) in prior season?
cumulative_days_injured	Total career injury days across all prior seasons

Group 3 — Career Workload:

Feature	Description
cumulative_minutes_played	Total career minutes across all prior seasons
cumulative_games_played	Total career games across all prior seasons
minutes_per_game_prev_seasons	Average minutes per game in prior seasons
avg_games_per_season_prev_seasons	Average games played per prior season

4. EXPLORATORY DATA ANALYSIS

4.1 Demographic Findings

Age versus High Injury Rate:

Age Group	High Injury Rate
Under 22	66.3%
22–25	71.0%
26–29	71.2%
30–34	72.5%
35+	78.7%

Clear clinical pattern: older players have progressively higher injury rates. The non-linear nature is important — the linear Pearson correlation with age is only +0.065, but tree-based models capture the threshold effect (particularly 30+ years) that linear models miss.

Position versus High Injury Rate:

Position	High Injury Rate
Goalkeeper	72.2%
Defender	70.8%
Midfielder	69.8%
Forward	69.2%

Only 3pp spread across positions — effectively negligible. Position contributes near-zero predictive value (Pearson $r \approx -0.020$).

Prior Injury History — The Most Powerful Signal:

Prior Injury Status	High Injury Rate	Mean Days Injured
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Had significant prior injury	82.2%	109.5 days
No significant prior injury	67.4%	71.3 days

Players with a significant prior injury are 14.8 percentage points more likely to sustain another significant injury. Consistent with 40+ years of sports medicine evidence: prior injury is the single strongest risk factor for future injury.

4.2 Correlation Analysis

Feature	Pearson r	Interpretation
cumulative_days_injured	+0.257	Strongest predictor — career injury burden
avg_days_injured_prev_seasons	+0.145	Prior injury rate
season_days_injured_prev_season	+0.145	Last season's injury burden
significant_injury_prev_season	+0.136	Binary prior injury flag
cumulative_games_played	+0.130	Career exposure
cumulative_minutes_played	+0.120	Career workload
age	+0.065	Weak linear but strong non-linear
pace, physic, bmi, height, weight	< 0.037	Near-zero — physical attributes irrelevant

Critical finding: Physical attributes have near-zero linear correlation with injury risk. Injury prediction is driven by injury history, not physique. This is deeply consistent with clinical evidence and confirmed by SHAP analysis.

4.3 Missing Values — Two Distinct Types

Type 1 — Structural NaN (first-season players): ~604 NaN values across history columns. These are not errors — a debut player genuinely has zero prior injury days, zero cumulative minutes. Correct fill: 0. Using median would fabricate injury history.

Type 2 — Data quality NaN (FIFA attributes): 95 missing pace/physic values. Correct fill: column median (pace=71.0, physic=72.3). Represents 'typical mid-range player' — reasonable when specific data unavailable. Impact minimal as these features have near-zero predictive power.

4.4 Feature Leakage Classification

Post-season leakage (7 columns dropped):

- `season_days_injured` — this IS the target itself
- `total_days_injured` — includes current season, revealing the answer
- `season_minutes_played`, `season_games_played`, `season_matches_in_squad` — current-season playing time is a consequence of injury, not a predictor
- `total_minutes_played`, `total_games_played` — cumulative totals including current season

This is consequential leakage — where a feature is a consequence of the target. A player injured for 3 months naturally played fewer minutes. Keeping `season_minutes_played` allows the model to reverse-engineer the target, not predict it.

5. PREPROCESSING

5.1 Pipeline

Step	Action	Before	After
1	Load raw data	—	(1,301, 30)
2	Construct High_Injury target (≥ 28 days)	Continuous days	Binary 0/1
3	Drop identifiers (<code>p_id2</code> , <code>dob</code> , <code>nationality</code> , etc.)	(1,301, 30)	Fewer cols
4	Drop 7 post-season leakage columns	—	(1,301, 20)
5	Zero-fill 8 history columns for first-season players	4,844 NaN	0
6	Median-impute <code>pace</code> (71.0) and <code>physic</code> (72.3)	95 NaN	Median
7	Mode-impute 2 missing <code>position_numeric</code>	2 NaN	1.0 (DEF)
8	Final validation	—	(1,301, 18), 0 missing
9	Save to <code>processed/</code>	—	<code>player_injuries_processed.csv</code>

5.2 Why Binary Rather Than Regression?

Predicting exact injury days (e.g., 'this player will be out for 47 days') is unreliable and clinically unhelpful. Enormous biological variance means two players with identical pre-season profiles can have very different outcomes. The binary formulation asks the more answerable and more actionable question: 'Is this player at high or low risk?' Medical and coaching staff can act on 'High Risk' with adjusted training loads and squad contingencies.

5.3 scale_pos_weight for This Dataset

Unlike Cycle 2 where `scale_pos_weight > 1` upweighted the rare positive (goal) class, here the positive class (High Injury) is the MAJORITY (70.2%). The minority class is Low Injury (class 0). Setting `scale_pos_weight = 0.42` ($= 388 \text{ Low} / 913 \text{ High} \approx 0.42$) downweights the High Injury class during training, giving the model more reason to attend to Low Injury examples.

6. MODELLING

6.1 Models and Baseline Results (Random Split)

Model	AUC-ROC	Accuracy	HI Recall	LI Recall	Note
Dummy	0.5000	70.11%	1.00	0.00	Always predicts High Injury
Logistic Regression	0.6220	55.17%	0.52	0.62	Best AUC despite lowest accuracy
Random Forest	0.5916	72.03%	0.97	0.04	Majority class collapse
XGBoost	0.6179	65.52%	0.78	0.34	Competitive with LR
LightGBM (BEST)	0.6355	67.43%	0.80	0.36	Best baseline

6.2 Random Forest Failure Explained

Random Forest achieves the highest accuracy (72.03%) but near-random AUC (0.5916). Despite `class_weight='balanced'`, it predicts High Injury for 97% of all players. This reveals the fundamental limitation of bagged ensembles on moderately imbalanced data: each individual tree develops slight biases toward the majority class. When 200 trees vote, small biases compound into an overwhelming majority-class vote. The model has essentially learned the Dummy Classifier strategy with extra computational steps.

6.3 Why AUC 0.68 is Acceptable

AUC 0.68 may appear modest compared to Cycle 2's 0.8183. The difference is fundamental: xG prediction depends on physics — near-deterministic relationships between shot position and goal probability. Injury prediction depends on biology, random contact events, and unobservable factors (training load, sleep quality, genetic predisposition). Published sports medicine injury prediction models using pre-season data consistently achieve AUC 0.65–0.75. The Cycle 3 model's 0.6800 is competitive with the state of the art.

7. HYPERPARAMETER TUNING

7.1 Random Split Tuning Results

Model	CV AUC	Test AUC	CV-Test Gap	Verdict
Tuned XGBoost	0.6584	0.6558	0.0026	WINNER — excellent generalisation
Tuned Random Forest	0.6411	0.6169	0.0241	Overfitting
Tuned LightGBM	0.6619	0.6273	0.0346	Regression vs baseline — overfitting

Important finding: Tuning decreased LightGBM's performance (0.6355 → 0.6273). num_leaves=50 on only 1,040 training rows creates over-specialised boundaries that do not generalise. This demonstrates a critical ML principle: tuning does not always improve performance. Small datasets relative to model complexity make over-fitting in the search space likely.

7.2 Best XGBoost Parameters (Random Split)

Parameter	Value	Rationale
n_estimators	100	Fewer trees — appropriate for small dataset
max_depth	4	Moderate depth prevents overfitting
learning_rate	0.2	Faster learning with fewer trees
subsample	1.0	Full data per tree

min_child_weight	5	Conservative — prevents overly specific splits
scale_pos_weight	0.4247	Near-exact computed ratio — already optimal

8. CHRONOLOGICAL VS RANDOM SPLIT

8.1 A Fundamentally Different Pattern from Cycle 1

In Cycle 1, chronological splitting reduced all model performances (−0.22pp to −4.17pp) — revealing leakage where future match information contaminated random-split training. In Cycle 3, the OPPOSITE happens. All four real models IMPROVE under chronological splitting:

Model	Random AUC	Chronological AUC	Delta
Logistic Regression	0.6220	0.6322	+0.0102
Random Forest	0.5916	0.6428	+0.0512
XGBoost	0.6179	0.6539	+0.0360
LightGBM	0.6355	0.6568	+0.0213

8.2 Why Chronological Split Improves Performance Here

Three explanations work together:

1. No temporal leakage in the random split. Unlike Cycle 1 (where team quality evolved season by season), pre-season biographical features (age, injury history, FIFA ratings) from 2016 are genuinely independent of what happens in 2020. There is no future information to remove.
2. The model learns to predict a harder-to-deceive future season. Training on 2016–2019 provides complete career progression data for players who appear in the 2020 test set at an advanced career stage. The chronological structure creates a natural forward-projection task that age and cumulative injury history are well-suited to support.
3. Random Forest's dramatic improvement (+0.0512). In the random split, shuffled seasons provide no temporal coherence. In the chronological split, training data is ordered correctly — players' career arcs build progressively from 2016 to 2019. This structured order helps the forest find splits recognising career progression patterns.

8.3 Chronological Tuning Results — Final Model

Model	Random AUC	Chrono AUC	Delta
Logistic Regression	0.6220	0.6263	+0.0043
Tuned XGBoost	0.6558	0.6723	+0.0165
Tuned Random Forest	0.6169	0.6668	+0.0499
Tuned LightGBM (WINNER)	0.6273	0.6800	+0.0527

Final deployed model: Tuned LightGBM (Chronological) — AUC 0.6800. Trained on 2016–2019 seasons (956 rows), tested on 2020 season only (242 rows). The same parameters that caused overfitting on the random split succeed on the chronological split because training data is now temporally coherent — the model finds genuine career-arc patterns that generalise to the 2020 test season.

8.4 Best LightGBM Parameters (Chronological)

Parameter	Value
n_estimators	200
learning_rate	0.01
max_depth	3
num_leaves	50
subsample	0.8
colsample_bytree	0.8
min_child_samples	20
scale_pos_weight	1.0

9. EVALUATION

9.1 Clinical Interpretation of Metrics

High Injury Recall (Sensitivity): What fraction of genuinely high-risk players does the model correctly flag? Missing a high-risk player (false negative) means they are injured without intervention — clinically costly.

Low Injury Recall (Specificity): What fraction of low-risk players does the model correctly clear? A system flagging every player as high-risk has zero specificity, wasting physiotherapy resources unnecessarily.

The medical asymmetry: A false negative is clinically costlier than a false positive. A model with high recall (even at some precision cost) is preferred — the tuned LightGBM's emphasis on injury history naturally produces this profile.

9.2 Strengths and Weaknesses

Strengths:

- **Medical validity:** Feature importance hierarchy precisely matches sports medicine evidence base.
- **Temporal integrity:** Training on 2016–2019, testing on 2020 — genuine forward-projection evaluation.
- **Two-type NaN strategy:** Structurally distinct NaN types handled with different appropriate strategies.
- **Actionable explanations:** SHAP waterfall plots provide clinically defensible per-player explanations.

Weaknesses:

- **No training load data:** GPS tracking, session intensity, and match congestion data are the primary causal injury drivers — completely absent.
- **Small dataset:** 1,301 player-seasons across 604 players is very small for medical prediction.
- **No injury type distinction:** An ACL rupture and hamstring strain are both labelled High Injury despite different mechanisms and prevention strategies.
- **Class collapse risk:** Even tuned Random Forest shows large CV-to-test gaps, suggesting instability without more data.

10. EXPLAINABILITY (SHAP)

10.1 Why Explainability is Critical in Sports Medicine

In Cycles 1 and 2, SHAP was used primarily to validate the model and power dashboard visualisations. In Cycle 3, explainability takes on an additional dimension: medical ethics and accountability.

A club's medical staff using a risk scoring system has a duty of care to players. If the model recommends modified training protocols or flags a player for early intervention, the medical team must be able to explain WHY to the player, the manager, and — in the event of a dispute — to legal or regulatory bodies. A black-box system that says 'Player X has 73% High Injury risk' without explanation is clinically unacceptable. SHAP provides the per-player, feature-level explanations required for clinical accountability.

10.2 Global Feature Importance

Rank	Feature	SHAP Magnitude	Interpretation
1	cumulative_days_injured	Highest	Total career injury burden — dominant predictor
2	avg_days_injured_prev_seasons	High	Per-season injury rate independent of career length
3	age	High	Non-linear threshold effect — particularly 30+ years
4	significant_injury_prev_season	Moderate	Binary prior injury — 'bounce-back' effect
5	season_days_injured_prev_season	Moderate	Granular recent injury burden
6–7	cumulative_games/minutes_played	Moderate	Career exposure and accumulated physical stress
8–9	minutes_per_game, bmi	Low	Marginal contributions
10–17	Physical attributes (height, weight, pace, physic, fifa_rating)	Near-zero	No predictive value — confirms clinical evidence

10.3 Medical Validation

The SHAP ranking maps perfectly onto established sports medicine literature:

- Prior injury history dominates (top 4 features) → consistent with 'prior injury as #1 risk factor' (Drawer & Fuller, 2001, and 20+ years of subsequent research)
- Age contributes non-linearly → consistent with age-related tissue degeneration
- Career workload → consistent with accumulated training load and tissue fatigue
- Physical attributes near-zero → consistent with clinical findings that raw measurements are poor independent predictors when injury history is known

Limitation of AUC 0.6800: The model cannot predict injury with high accuracy from pre-season data alone because the primary causal driver — in-season training load, biomechanical stress exposure — is not in the dataset. AUC 0.68 is the genuine ceiling of what pre-season biographical features can achieve. It should be used as a screening and risk stratification tool to supplement clinical assessment, not replace it.

11. DEPLOYMENT

11.1 Saved Artefacts

- cycle3_best_model.pkl — Tuned LightGBM (AUC 0.6800, trained on 2016–2019)
- cycle3_scaler.pkl — StandardScaler fitted on training seasons only
- cycle3_feature_cols.pkl — 17 feature names in exact required order

11.2 Streamlit Dashboard

The Cycle 3 dashboard is a pre-season player risk assessment tool: (1) User enters player details (age, height, weight, FIFA ratings, prior injury history, career workload); (2) Dashboard sends 17 features to the FastAPI endpoint; (3) Model returns injury risk probability; (4) Dashboard displays: risk score (0–1), risk category (High/Low Risk), colour-coded risk gauge, and SHAP waterfall explaining which of the player's features are driving their specific risk score.

11.3 Full Pipeline

```
User enters player pre-season profile
→ Dashboard assembles 17 features
→ POST request → FastAPI endpoint
→ Load pkl files → Align features
→ Apply StandardScaler transformation
→ LightGBM.predict_proba(scaled_input)[0][1]
→ Return injury risk probability + classification
→ Streamlit renders: risk gauge + SHAP waterfall
```

12. CONCLUSION

Cycle 3 delivered a complete, production-ready player injury risk pipeline. 1,301 player-season records were preprocessed into a clean 17-feature dataset with zero missing values and zero data leakage. The clinically meaningful 28-day binary threshold was applied. Five models were compared under both random and chronological conditions, revealing the counterintuitive finding that chronological splitting improves performance — a direct consequence of career-arc temporal coherence. Tuned LightGBM (Chronological) achieves AUC 0.6800, consistent with published sports medicine injury prediction benchmarks. SHAP confirmed a medically validated feature ranking.

13. FUTURE IMPROVEMENTS

- Integrate GPS and session load data: Daily session distance, sprint counts, and fatigue indices — primary causal drivers, likely pushing AUC above 0.75.
- Injury type classification: Extend to multi-class predicting muscular, ligament, bone, and overuse injuries with type-specific prevention strategies.
- Player-specific longitudinal modelling: RNNs or Hidden Markov Models to capture season-to-season career dynamics.
- Incorporate match congestion features: European competition, international duty, and consecutive matches are publicly available and injury-relevant.
- Severity-weighted target: Train on injury days (regression) or severity quartile (ordinal) for more actionable predictions.
- Real-time probability updating: Re-run the model each matchweek with updated rolling workload statistics for dynamic in-season risk scores.